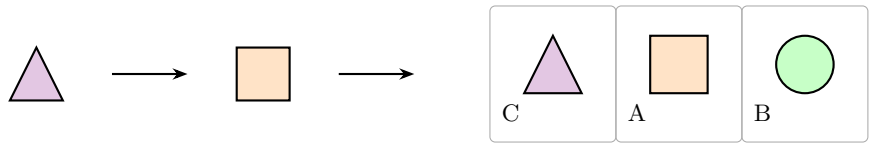
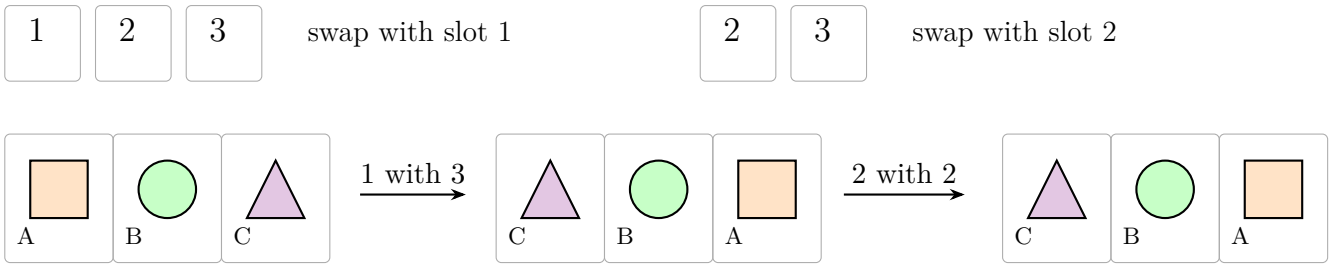
Mix the three picture cards in a cup. Each card has the same chance. Draw the first without looking and keep it out. Draw the second from the two left. Put the last card at the end. Return all cards for a new row.



Example: choose triangle, then square; circle is left.

Problem 1: Does this chooser give every order of A, B, C the same chance? Settle the question using every possible choice story.

Start A B C. Mix tickets 1, 2, 3 in the cup; draw and swap its slot with slot 1. Empty the cup, then mix tickets 2, 3; draw and swap its slot with slot 2. At each draw, all tickets in the cup have equal chances. Reset both sets for every new row. A slot may swap with itself.



Problem 2: Does this two-swap rule give every order the same chance? Find all its choice stories and their final orders.

Problem 3: Start with C B A instead. Explain whether the chance of each final order changes, without listing every story again.

